Movie Recommendation System

Content-Based Filtering Using Cosine Similarity

Technical Report

Python | Pandas | Scikit-learn

1. Project Overview

This project implements a Content-Based Movie Recommendation System using Python. The system analyses movie genres to find and suggest films that are similar to a user-provided title. By converting genre data into numerical vectors and applying cosine similarity, the recommender can identify movies that share comparable genre profiles.

The system is designed with two operational modes — a demo mode for quick automated testing and an interactive mode that accepts user input — making it both easy to evaluate and practical for end-users.

2. Objectives

- Build a lightweight, genre-based movie recommendation engine without requiring user history or ratings.
- Apply CountVectorizer to transform textual genre data into machine-readable feature vectors.
- Use Cosine Similarity to compute pairwise similarity scores across all movies in the dataset.
- Return the top 5 most similar movies for any given input title.
- Provide both an interactive CLI mode and an automated demo mode for flexible usage.

3. Technologies Used

The following technologies and libraries power the recommendation system:

Technology	Purpose	Version
Python	Core programming language	3.8+
Pandas	Data manipulation & DataFrame	1.x+
Scikit-learn	ML toolkit: CountVectorizer, Cosine Similarity	1.x+
CountVectorizer	Genre text → numerical feature matrix	Scikit-learn
Cosine Similarity	Measure movie-to-movie similarity	Scikit-learn

4. Dataset

A curated dataset of 34 movies is embedded directly in the script, modelled after the MovieLens dataset structure. Each record contains a movie title and a space-separated genre string. The dataset spans a wide range of genres including Action, Drama, Comedy, Animation, SciFi, Thriller, Romance, and War.

4.1 Sample Movie Records

The table below shows a representative selection from the dataset:

Movie Title	Genres
Toy Story	Animation, Children, Comedy, Fantasy
The Dark Knight	Action, Crime, Drama, Thriller
Inception	Action, Crime, Drama, Mystery, SciFi, Thriller
Interstellar	Adventure, Drama, SciFi
The Matrix	Action, SciFi, Thriller
GoodFellas	Crime, Drama
Pulp Fiction	Crime, Drama, Thriller
Titanic	Drama, Romance
Forrest Gump	Comedy, Drama, Romance

Movie Title	Genres
Jurassic Park	Action, Adventure, SciFi, Thriller

Full dataset includes 34 titles spanning genres such as Children, Musical, Fantasy, Mystery, Horror, and Adventure in addition to the genres shown above.

5. How It Works

The recommendation pipeline follows five sequential steps from raw data to a ranked list of suggestions:

Step	Phase	Description
1	**Data Preparation**	Movie titles and genres are loaded into a Pandas DataFrame for structured access.
2	**Feature Extraction**	CountVectorizer converts genre strings into a binary numerical matrix (1 = genre present, 0 = absent).
3	**Similarity Computation**	Cosine Similarity calculates pairwise similarity scores across all movies (1.0 = identical, 0.0 = no match).
4	**Recommendation**	User provides a movie title; system retrieves scores, sorts them, and returns the top 5 most similar movies.
5	**Output**	Results displayed in a clean ranked table with title, genres, and similarity score.

5.1 Feature Extraction Detail

CountVectorizer tokenises each movie's genre string (e.g., "Action Crime Thriller") and builds a binary matrix where each column corresponds to a unique genre. A cell value of 1 means the movie belongs to that genre; 0 means it does not. This produces a sparse matrix of shape (34 movies × N unique genres).

5.2 Cosine Similarity Formula

Cosine similarity measures the cosine of the angle between two genre vectors:

$$\text{similarity}(A, B) = (A \cdot B) / (||A|| \times ||B||)$$

A score of 1.0 means the two movies share identical genre profiles; 0.0 means they share no genres at all. Scores between 0 and 1 represent partial genre overlap.

6. System Features

- Case-insensitive search — input matching ignores capitalisation (e.g., 'toy story' matches 'Toy Story').
- Similarity score display — every recommendation shows its computed cosine similarity score.
- Top-5 recommendations — the system returns the five most genre-similar movies.
- Interactive mode — a looping CLI interface lets users query multiple movies in one session.
- Demo mode — runs automatic recommendations for three preset movies with no user input needed.
- Error handling — invalid movie names produce a helpful message listing available options.
- Clean tabular output — results are rendered as a formatted Pandas DataFrame table.

7. How to Run

7.1 Prerequisites

Install the required Python libraries before running the script:

```
pip install pandas scikit-learn
```

7.2 Running the Script

```
python movie_recommender.py
```

7.3 Mode Selection

- Enter 1 to run Demo mode — automatically recommends movies for Toy Story, The Dark Knight, and Titanic.
- Enter 2 to run Interactive mode — type any movie from the dataset to receive personalised recommendations.
- In interactive mode, type quit, exit, or q to terminate the session.

8. Sample Output

Below is an example of the system output when querying 'The Dark Knight':

```
Input Movie  : The Dark Knight
Genres       : Action Crime Drama Thriller

Top 5 Recommended Movies:
  1. Pulp Fiction        Crime Drama Thriller       0.8165
  2. GoodFellas          Crime Drama                0.7071
  3. Heat                Action Crime Thriller      0.6667
  4. Inception           Action Crime Drama ...     0.6325
  5. Se7en               Crime Mystery Thriller     0.5774
```

9. Limitations & Future Scope

9.1 Current Limitations

- The dataset is small (34 movies) and manually curated; a production system would use the full MovieLens dataset (25M+ ratings).
- Recommendations are based solely on genres — plot, cast, director, and user ratings are not considered.
- No collaborative filtering — the system cannot leverage what similar users have watched.
- No persistent storage — recommendations are not saved between sessions.

9.2 Potential Enhancements

- Integrate the full MovieLens dataset (movies.csv + ratings.csv) for broader coverage.
- Add collaborative filtering (SVD, ALS) to complement content-based recommendations.
- Incorporate TF-IDF weighting to prioritise rarer genres over common ones.
- Build a web interface using Flask or Streamlit for a graphical user experience.
- Add movie poster display and TMDB API integration for rich metadata.

10. Conclusion

This Movie Recommendation System demonstrates a clean, efficient implementation of content-based filtering using fundamental machine learning techniques. By combining CountVectorizer for feature extraction with Cosine Similarity for similarity measurement, the system accurately identifies genre-aligned movie suggestions with minimal computational overhead.

The modular code structure — separating data preparation, feature engineering, similarity computation, and recommendation logic — makes it straightforward to extend with larger datasets, richer features, or alternative algorithms. This project serves as a solid foundation for building more sophisticated, production-grade recommendation engines.

End of Report